1 Foundations

Definition. A *permutation* of size n is an ordering of $1, 2, \ldots, n$; for example, 123, 321, and so on.

Lemma. There are $n!$ permutations of length n.

Proof. There are n ways to make the first choice, then $n-1$ ways to make the next choice, and so on.

Definition. A *binary string* of length n is an n-tuple made of 0s and 1s.

Example. The binary strings of length 2 are 00, 01, 10, and 11.

Lemma. There are 2^n binary strings of length n.

Proof. Every position has two choices. Making these choices in n positions gives 2^n strings.

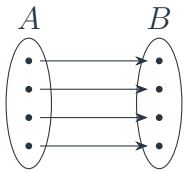
1.1 Bijections

The idea is to use things we already know to define new things.

Definition. Let A and B be sets. A function $f\colon A \to B$ is

- *injective* if $a \neq a'$ implies $f(a) \neq f(a')$ for all $a, a' \in A$; equivalently, $f(a) = f(a')$ implies $a = a'$;
- *surjective* if, for every $b \in B$, there is an $a \in A$ such that $f(a) = b$;
- *bijective* if it is both injective and surjective.

If A and B are finite and in bijection, then $|A| = |B|$.



A well-defined function $f\colon A \to B$ is a bijection if and only if there is a function $g\colon B \to A$ such that

$$f(g(b)) = b \quad (b \in B), \qquad g(f(a)) = a \quad (a \in A).$$

Here g is the *inverse* of f. The first identity gives surjectivity of f and the second gives injectivity. To prove that f is a bijection, we can either

1. show that it is injective and surjective, or
2. find its inverse.

1.2 Subsets

Definition. A *subset* of $[n] = \{1, \dots, n\}$ is a collection of some elements of $[n]$, possibly empty. The only subset of $[0] = \varnothing$ is $\varnothing$, the empty set.

Lemma. For $n \in \mathbb{N}$, there are 2^n subsets of $[n]$.

Proof. Every element is either picked or not picked, giving 2^n choices.

Bijective proof. Fix n. Let A be the set of subsets of $[n]$, and let B be the set of binary strings of length n. Define $f\colon A \to B$ by

$$f(S) = \sigma_1\sigma_2\cdots\sigma_n, \qquad \sigma_i = \begin{cases} 0, & i \notin S, \\ 1, & i \in S. \end{cases}$$

We need to show that f is well-defined, injective, and surjective.

1. For $S \in A$, the string $f(S)$ always has length n and every entry is 0 or 1, so $f(S) \in B$.
2. If $f(S) = f(T)$, then $i \in S$ if and only if $i \in T$, so $S = T$.
3. Given $\sigma_1 \cdots \sigma_n \in B$, let $S = \{i : \sigma_i = 1\}$. Then $S \in A$ and $f(S) = \sigma_1 \cdots \sigma_n$ by definition.

Thus $|A| = |B| = 2^n$.

Lemma. There are $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$ subsets of $[n]$ with k elements, for $n \in \mathbb{N}$ and $0 \le k \le n$.

Proof. There are

$$n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!}$$

ways to choose a tuple of length k using distinct elements of $[n]$. A subset of size k is such a tuple with the order disregarded. Since there are $k!$ ways to order its elements, there are

$$\frac{1}{k!}\frac{n!}{(n-k)!} = \binom{n}{k}$$

subsets of $[n]$ of size k.

1.3 Multisets

Definition. A *multiset* of size n with t types is a collection of n elements in $\{1, \ldots, t\}$ in which elements can repeat. Equivalently, it is a sequence $(m_1, \ldots, m_t) \in \mathbb{N}^t$ with $m_1 + \cdots + m_t = n$.

The two descriptions are related by letting m_i be the number of elements of type i. Here, and throughout these notes, $0 \in \mathbb{N}$.

Theorem. For $n, t \in \mathbb{N}$ and $t \geq 1$, there are $\binom{n+t-1}{t-1}$ multisets of size n with t types.

Idea of proof. For example, $m_1 + m_2 + m_3 = 5$ with $(m_1, m_2, m_3) = (1, 2, 2)$ can be represented by five balls and two dividing lines.



We can move the lines around. In general, we have n balls and $t-1$ lines to divide them. Choose the $t-1$ positions for the lines among $n+t-1$ positions, giving $\binom{n+t-1}{t-1}$ possibilities.

1.4 Combinatorial proofs

Bijections can give *combinatorial proofs* of identities $\mathrm{LHS} = \mathrm{RHS}$:

1. Find sets A and B such that $\mathrm{LHS} = |A|$ and $\mathrm{RHS} = |B|$.
2. Prove that A and B are in bijection.
3. Conclude that $\mathrm{LHS} = |A| = |B| = \mathrm{RHS}$.

A special case is to count one set in two ways.

Example. Give a combinatorial proof that $\binom{n}{k} = \binom{n}{n-k}$ for $n, k \in \mathbb{N}$ and $0 \leq k \leq n$.

Proof. Let

$$A = \{\text{subsets of } [n] \text{ of size } k\}, \qquad B = \{\text{subsets of } [n] \text{ of size } n-k\}.$$

Then $|A| = \binom{n}{k}$ and $|B| = \binom{n}{n-k}$. Define $f\colon A \to B$ by $f(S) = [n] \setminus S$, the complement of S in $[n]$.

1. If $|S| = k$, then $|f(S)| = |[n] \setminus S| = n - k$, so f is well-defined.
2. Taking the complement twice gives $[n] \setminus ([n] \setminus S) = S$, so the same complement operation is the inverse.

Thus f is a bijection and $\binom{n}{k} = |A| = |B| = \binom{n}{n-k}$.

Example. Prove that $2^n = \sum_{k=0}^{n} \binom{n}{k}$.

Proof. Let A be the set of all subsets of $[n]$, so $|A| = 2^n$. Write A as the disjoint union

$$A = A_0 \cup A_1 \cup \cdots \cup A_n, \qquad A_k = \{\text{subsets of } [n] \text{ of size } k\}.$$

Disjointness ensures that no subset is counted twice. Therefore

$$\begin{aligned} 2^n &= |A| \\ &= \left| \bigcup_{k=0}^{n} A_k \right| \\ &= \sum_{k=0}^{n} |A_k| \\ &= \sum_{k=0}^{n} \binom{n}{k}. \end{aligned}$$

Common objects in combinatorial proofs include subsets, permutations, binary strings, multisets, and lattice paths.

Definition. A *lattice path* is a sequence of steps that starts and ends at specified points in $\mathbb{Z}^d$.

Example. Find the number of lattice paths in $\mathbb{Z}^2$ that start at $(0,0)$ and end at $(a,b) \in \mathbb{N}^2$, using the steps $N = (0,1)$ and $E = (1,0)$.

A walk ending at (a,b) must have a east steps and b north steps, for a total of $a+b$ steps. Choose the a east-step positions; the remaining steps are uniquely determined. Hence there are $\binom{a+b}{a}$ paths.

2 A general framework for enumeration

Definition. A *combinatorial class* is a set A together with a size or weight function $w\colon A \to \mathbb{N}$ such that there are finitely many objects of each size. In other words,

$$A_n = w^{-1}(n) = \{a \in A : w(a) = n\}$$

is finite for every $n \in \mathbb{N}$. Typically we write $w(a) = |a|$.

Remark. The entire set A need not be finite; only each set of objects of a fixed weight must be finite.

Example. Let $B = \{\varepsilon, 0, 1, 00, 01, 10, 11, \ldots\}$ be the set of binary strings, where ε is the empty string.

- If $|s|$ is the length of s, then $(B, |\cdot|_B)$ is a combinatorial class.
- If $|s|$ is the number of 0s in s, this is not a combinatorial class: $B_0 = \{\varepsilon, 1, 11, \ldots\}$ is infinite.

We consider different weights because we may want to count other properties, such as the number of 1s or 0s.

Definition. A *bijection between classes* $(A, |\cdot|_A)$ and $(B, |\cdot|_B)$ is a bijection $f\colon A \to B$ that respects size:

$$|a|_A = |f(a)|_B \qquad \text{for all } a \in A.$$

The *counting sequence* of a class $(A, |\cdot|)$ is defined by

$$a_n = |A_n| = \text{number of objects of size } n.$$

For binary strings with length as size, $b_n = 2^n$.

2.1 Generating functions and formal series

Definition. The *generating function* (or *generating series*) of a sequence (a_n) is

$$\sum_{n\geq 0} a_n z^n = a_0 + a_1 z + a_2 z^2 + \cdots.$$

The generating function of a class $(A, |\cdot|)$ is the generating function of its counting sequence.

Example. The generating function for the class B of binary strings is

$$B(z) = \sum_{n\geq 0} b_n z^n = \sum_{n\geq 0} 2^n z^n.$$

This is a *formal series*. Analytically, $\sum_{n\geq 0} 2^n z^n$ converges for $|z| < \frac{1}{2}$, whereas $\sum_{n\geq 0} n! z^n$ diverges unless $z = 0$.

For formal series, define addition and multiplication by

$$\sum_{n\geq 0} a_n x^n + \sum_{n\geq 0} b_n x^n = \sum_{n\geq 0} (a_n + b_n) x^n,$$

$$\left(\sum_{n\geq 0} a_n x^n\right)\left(\sum_{n\geq 0} b_n x^n\right) = \sum_{n\geq 0}\left(\sum_{k=0}^{n} a_k b_{n-k}\right) x^n.$$

The usual algebraic properties still hold formally. We write $\mathbb{Q}[[x]]$ for the ring of formal power series with rational coefficients, and $\mathbb{Q}[x]$ for the ring of polynomials.

Example. Consider $F(x) = 1 - x$. Does there exist a series $G(x) = \sum_{n\geq 0} g_n x^n$ such that $F(x)G(x) = 1$?

Let $G(x) = \sum_{n\geq 0} x^n$. Then

$$\begin{aligned} F(x)G(x) &= (1-x)\left(\sum_{n\geq 0} x^n\right) \\ &= \sum_{n\geq 0} x^n - x\sum_{n\geq 0} x^n \\ &= 1 + \sum_{n\geq 1} x^n - \sum_{n\geq 1} x^n = 1. \end{aligned}$$

Thus $\dfrac{1}{F(x)} = G(x) = \dfrac{1}{1-x}$ as formal series.

Example. If $F(x) = x + x^2$, then formally factoring gives

$$\frac{1}{F(x)} = \frac{1}{x}\frac{1}{1+x} = \frac{1}{x}\sum_{n\geq 0}(-1)^n x^n.$$

This contains an x^{-1} term, so it is not a formal power series. No $G(x) \in \mathbb{Q}[[x]]$ can satisfy $G(x)F(x) = 1$.

Lemma. If $F(x) = \sum_{n\geq 0} f_n x^n$, then $1/F(x)$ exists in $\mathbb{Q}[[x]]$ if and only if $f_0 \neq 0$.

Lemma. The formal substitution $F(G(x))$ is defined if and only if F is a polynomial or $g_0 = 0$, where $G(x) = \sum_{n\geq 0} g_n x^n$.

Indeed, in $F(G(x)) = \sum_{n\geq 0} f_n G(x)^n$, each coefficient must receive only finitely many contributions.

2.2 Examples of generating functions

Example. Find the generating function for m_n, the number of ways to make change for n cents using 1, 5, 10, and 25 cent coins.

Let $M = \{(a,b,c,d) \in \mathbb{N}^4\}$ and define $|(a,b,c,d)| = a + 5b + 10c + 25d$. Then

$$M(x) = \sum_{\sigma\in M} x^{|\sigma|} = \sum_{(a,b,c,d)\in M} x^{a+5b+10c+25d}$$

$$= \left(\sum_{a\geq 0} x^a\right)\left(\sum_{b\geq 0} x^{5b}\right)\left(\sum_{c\geq 0} x^{10c}\right)\left(\sum_{d\geq 0} x^{25d}\right)$$
$$= \frac{1}{(1-x)(1-x^5)(1-x^{10})(1-x^{25})}.$$

For example, substituting $G(x) = x^5$ into $F(x) = 1/(1-x)$ is valid since $G(0) = 0$. Computing coefficients gives

$$M(x) = 1 + x + x^2 + \cdots + 242x^{100} + \cdots.$$

There are 242 ways to make change for one dollar. Asymptotically, $m_n \sim n^3/7500$.

Example. The generating function for the sum of two standard dice is

$$\begin{aligned} D(x) &= (x + x^2 + x^3 + x^4 + x^5 + x^6)^2 \\ &= (x + 2x^2 + 2x^3 + x^4)(x + x^3 + x^4 + x^5 + x^6 + x^8). \end{aligned}$$

The coefficients in each factor sum to 6. Thus two dice with faces

$$(1, 2, 2, 3, 3, 4) \qquad \text{and} \qquad (1, 3, 4, 5, 6, 8)$$

give the same probability of rolling each possible sum as two standard dice.

Remark. Recall that for a class $\mathcal{C}$,

$$C(x) = \sum_{n\geq 0} c_n x^n = \sum_{\sigma\in\mathcal{C}} x^{|\sigma|}.$$

The reciprocal $1/F(x)$ exists as a formal series exactly when $F(0) \neq 0$. A composition $G(F(x))$ is formally defined if and only if G is a polynomial or $F(0) = 0$.

2.3 Binomial theorems

Theorem (Binomial theorem)**.** If $n \in \mathbb{N}$, then

$$(1 + x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

Theorem (Negative binomial theorem)**.** If t is a positive integer, then

$$\frac{1}{(1-x)^t} = \sum_{n\geq 0} \binom{n+t-1}{t-1} x^n.$$

Proof. The right-hand side is the generating function for multisets with t types. Thus

$$\sum_{n\geq 0}\binom{n+t-1}{t-1}x^n = \sum_{(m_1,\ldots,m_t)\in\mathbb{N}^t} x^{m_1+\cdots+m_t}$$
$$= \left(\sum_{m_1\geq 0} x^{m_1}\right)\cdots\left(\sum_{m_t\geq 0} x^{m_t}\right) = \frac{1}{(1-x)^t}.$$

Definition. For $k\in\mathbb{N}$ and $\alpha\in\mathbb{C}$, define

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}, \qquad \binom{\alpha}{0} = 1.$$

Remark. For integers $\alpha\geq k$, this agrees with $\dfrac{\alpha!}{k!(\alpha-k)!}$. For arbitrary complex α, we use the falling-product definition above; ordinary factorials are not defined for all complex arguments.

Theorem (Generalized binomial theorem)**.** For any $\alpha\in\mathbb{C}$,

$$(1+x)^\alpha = \sum_{n\geq 0}\binom{\alpha}{n}x^n.$$

From the negative binomial theorem,

$$\binom{n+t-1}{t-1} = (-1)^n\binom{-t}{n}.$$

2.4 Coefficient extraction

We write $[x^n]\left(\sum_{k\geq 0} c_k x^k\right) = c_n$. In particular, $[x^0]F(x) = f_0 = F(0)$. The rules for coefficient extraction are

$$[x^n]\lambda A(x) = \lambda[x^n]A(x),$$
$$[x^n](A(x)+B(x)) = [x^n]A(x) + [x^n]B(x),$$
$$[x^n]F(ax) = a^n[x^n]F(x),$$
$$[x^n]x^kF(x) = [x^{n-k}]F(x).$$

The last expression is zero when $k>n$. The scaling rule follows by writing $F(ax) = \sum_{n\geq 0} f_n a^n x^n$.

Example. Find $[x^n]\dfrac{1-\sqrt{1-4x}}{2x}$. This is the Catalan series, which is in bijection with many different classes of objects.

$$
\begin{aligned}
[x^n]\frac{1-\sqrt{1-4x}}{2x} &= \frac{1}{2}[x^{n+1}](1-\sqrt{1-4x}) \\
&= -\frac{1}{2}[x^{n+1}]\sqrt{1-4x} \\
&= -\frac{1}{2}(-4)^{n+1}[x^{n+1}](1+x)^{1/2} \\
&= -\frac{1}{2}(-4)^{n+1}\binom{1/2}{n+1} \\
&= -\frac{1}{2}(-4)^{n+1}\frac{\frac{1}{2}(\frac{1}{2}-1)\cdots(\frac{1}{2}-n)}{(n+1)!} \\
&= \frac{2^n(-2)^n(\frac{1}{2}-1)(\frac{1}{2}-2)\cdots(\frac{1}{2}-n)}{(n+1)!} \\
&= \frac{(1)(3)(5)\cdots(2n-1)\quad \underbrace{2^n n!}_{2\cdot 4\cdots(2n)}}{(n+1)!\,n!} \\
&= \frac{(2n)!}{n!\,n!}\frac{1}{n+1} = \frac{1}{n+1}\binom{2n}{n}.
\end{aligned}
$$

3 Constructions of combinatorial classes

The idea is to build complicated classes from simple ones and translate these constructions into generating functions.

3.1 Two basic classes

1. The *atomic class* $\mathcal{Z} = \{\bullet\}$ has a single object of size 1. Its generating function is z.
2. The *neutral class* $\mathcal{E} = \{\varepsilon\}$ has a single object of size 0. Its generating function is 1.

3.2 Sum

Definition. Let $(A, |\cdot|_A)$ and $(B, |\cdot|_B)$ be combinatorial classes. The *sum* $C = A + B$ is the class whose objects are the disjoint union of the objects in A and B, with size

$$
|\sigma|_C = \begin{cases} |\sigma|_A, & \sigma \in A, \\ |\sigma|_B, & \sigma \in B. \end{cases}
$$

Example. If $\mathcal{Z}$ is the atomic class, then $\mathcal{Z} + \mathcal{Z}$ consists of two distinct atoms of size 1.

Lemma. If $C = A + B$, then $C(z) = A(z) + B(z)$.

Proof. An object of size n in C is an object of size n in A or in B. Because the union is disjoint, $c_n = a_n + b_n$. Thus

$$\begin{aligned} C(z) &= \sum_{n\geq 0} c_n z^n = \sum_{n\geq 0} (a_n + b_n) z^n \\ &= \sum_{n\geq 0} a_n z^n + \sum_{n\geq 0} b_n z^n = A(z) + B(z). \end{aligned}$$

3.3 Product and powers

Definition. The *product* $C = A \times B$ is the class of all pairs

$$\{(\alpha, \beta) : \alpha \in A,\ \beta \in B\}, \qquad |(\alpha,\beta)|_C = |\alpha|_A + |\beta|_B.$$

Lemma. If $C = A \times B$, then $C(z) = A(z)B(z)$.

Proof.

$$\begin{aligned} C(z) &= \sum_{(\alpha,\beta)\in C} z^{|(\alpha,\beta)|} = \sum_{(\alpha,\beta)\in C} z^{|\alpha|_A + |\beta|_B} \\ &= \left(\sum_{\alpha\in A} z^{|\alpha|_A}\right)\left(\sum_{\beta\in B} z^{|\beta|_B}\right) = A(z)B(z). \end{aligned}$$

For any positive integer k, the *power*

$$A^k = \underbrace{A \times A \times \cdots \times A}_{k \text{ times}} = \{(\alpha_1, \ldots, \alpha_k) : \alpha_i \in A\}$$

has generating function $A(z)^k$.

3.4 Sequences

Definition. Let A be a combinatorial class with no objects of size 0. Its *sequence class* is

$$\mathrm{SEQ}(A) = \mathcal{E} + A + A^2 + \cdots = \bigcup_{k\geq 0} \{(\alpha_1, \ldots, \alpha_k) : \alpha_i \in A\}.$$

If A had an object of size 0, then $\mathrm{SEQ}(A)$ would have infinitely many objects of size 0.

Lemma. If $C = \mathrm{SEQ}(A)$, then

$$C(z) = 1 + A(z) + A(z)^2 + \cdots = \frac{1}{1 - A(z)}.$$

Example. Let B be the class of binary strings. Taking the atoms $\mathcal{Z}_0 = \{0\}$ and $\mathcal{Z}_1 = \{1\}$,

$$B = \mathrm{SEQ}(\mathcal{Z}_0 + \mathcal{Z}_1), \qquad B(z) = \frac{1}{1-(z+z)} = \frac{1}{1-2z}.$$

Definition. An *integer composition* of size n is a tuple of positive integers whose sum is n.

Example. The compositions of size 3 are (3), $(2,1)$, $(1,2)$, and $(1,1,1)$, corresponding to 3, $2+1$, $1+2$, and $1+1+1$. If order does not matter, the corresponding object is a partition. Let P be the class of positive integers, with $|n| = n$, and let C be the class of compositions. Then

$$C = \mathrm{SEQ}(P), \qquad P = \mathcal{Z} \times \mathrm{SEQ}(\mathcal{Z}).$$

The initial atom ensures that the integer is positive, because $\mathrm{SEQ}(\mathcal{Z})$ includes the empty object. Thus

$$P(z) = \frac{z}{1-z}, \qquad C(z) = \frac{1}{1-\frac{z}{1-z}} = \frac{1-z}{1-2z}.$$

For $n \geq 1$,

$$\begin{aligned}
[z^n]C(z) &= [z^n]\frac{1-z}{1-2z} \\
&= [z^n]\left(\frac{1}{1-2z} - \frac{z}{1-2z}\right) \\
&= 2^n - [z^{n-1}]\frac{1}{1-2z} \\
&= 2^n - 2^{n-1} = 2^{n-1}.
\end{aligned}$$

The restriction $n \geq 1$ is necessary; the empty composition gives $[z^0]C(z) = 1$.

Restricted sequence constructions give

$$\mathrm{SEQ}_{\leq r}(A) = \mathcal{E} + A + \cdots + A^r, \qquad C(z) = \frac{1-A(z)^{r+1}}{1-A(z)},$$

$$\mathrm{SEQ}_{\geq r}(A) = A^r + A^{r+1} + \cdots, \qquad C(z) = \frac{A(z)^r}{1-A(z)},$$

$$\mathrm{SEQ}_{\mathrm{even}}(A) = \mathcal{E} + A^2 + A^4 + \cdots, \qquad C(z) = \frac{1}{1-A(z)^2}.$$

3.5 Iterative and recursive constructions

There are two types of constructions: *iterative* and *recursive*.

Example. A rooted planar binary tree is either empty or has a root vertex followed by a possibly empty left subtree and a possibly empty right subtree. In particular, a single root vertex is allowed.



If B is the class of these trees, then

$$B = \mathcal{E} + \mathcal{Z} \times B \times B.$$

The four components mean: empty tree, root, left subtree, and right subtree. Therefore

$$B(z) = 1 + zB(z)^2, \qquad zB(z)^2 - B(z) + 1 = 0,$$

so $B(z)$ is one of the two solutions

$$\frac{1 \pm \sqrt{1-4z}}{2z}.$$

The plus sign gives

$$\frac{1+\sqrt{1-4z}}{2z} = \frac{2 - 2z + \cdots}{2z} = \frac{1}{z} - 1 + \cdots,$$

which is not a formal power series.

Therefore the binary-tree generating function is

$$B(z) = \frac{1-\sqrt{1-4z}}{2z} = 1 + z + 2z^2 + \cdots = \sum_{n \geq 0} \frac{1}{n+1}\binom{2n}{n} z^n.$$

This counts the possible binary trees of a given size; the source also notes its relevance to estimating the number of bits needed to encode a binary tree.

4 Binary strings and regular expressions

We have seen how to count all binary strings: $b_n = 2^n$ and

$$B(x) = \sum_{n \geq 0} b_n x^n = \sum_{n \geq 0} 2^n x^n = \frac{1}{1-2x}.$$

We now consider binary strings with restrictions.

4.1 Concatenation

We write strings as sequences of digits without commas: $(0,1,1,0) = 0110$.

Definition. The *concatenation* of $\sigma = \sigma_1\sigma_2\cdots\sigma_r$ and $\tau = \tau_1\tau_2\cdots\tau_s$ is

$$\sigma\tau = \sigma_1\sigma_2\cdots\sigma_r\tau_1\tau_2\cdots\tau_s.$$

- The empty string ε has size 0 and satisfies $\varepsilon\sigma = \sigma = \sigma\varepsilon$ for every string σ.
- We write $\sigma^k = \underbrace{\sigma\sigma\cdots\sigma}_{k \text{ times}}$.
- We say that σ is a *substring* of τ if σ occurs within τ with no gaps.

- The *concatenation product* of sets S and T of strings is $ST = \{\sigma\tau : \sigma \in S,\ \tau \in T\}$.
- We write $S^k = \underbrace{SS\cdots S}_{k \text{ times}}$, $S^0 = \{\varepsilon\}$, and $S^* = \bigcup_{k\geq 0} S^k$.

Remark. Concatenation can identify different pairs. For example,

$$\{1, 10\}\{1, 01\} = \{11, 101, 101, 1001\} = \{11, 101, 1001\},$$

whereas the Cartesian product consists of the four distinct pairs

$$\{(1, 1), (10, 1), (1, 01), (10, 01)\}.$$

Example.
- $\{0, 1\}^*$ is the set of all binary strings.
- $\{0, 1\}\{0, 1\}^*$ is the set of all nonempty binary strings.
- $(\{0, 1\}^2)^*$ is the set of all binary strings of even length.

A set of strings can be specified in more than one way. For example,

$$\{0, 1\}^* = \{\varepsilon, 0, 1\}^*$$

as sets of strings, but the expressions do not describe the same ways of producing a string: empty strings can be inserted in the second description. We need to consider how strings are produced, not just which strings are obtained.

4.2 Regular expressions and ambiguity

Definition. A *regular expression* is defined recursively by one of the following:

1. ε, 0, or 1;
2. $R \cup S$, where R and S are regular expressions;
3. RS, where R and S are regular expressions;
4. R^*, where R is a regular expression.

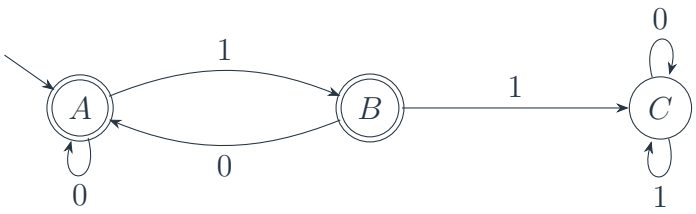
A regular expression R produces a set of strings $\mathcal{L}(R)$, called a *regular language*.

We use $0 \cup 1$ as shorthand for $\{0, 1\}$.

Example.

- $(00)(1 \cup 01)$ produces $\{001, 0001\}$.
- $(10)^5(111)$ produces $\{1010101010111\}$.
- $(0 \cup 1)^*$ produces all binary strings.

Regular languages are important in computer science because they are exactly the languages accepted by finite automata. For example, the following automaton accepts all binary strings with no two consecutive 1s. Double circles indicate accepting states.



Example. The expression $(1 \cup 10)(1 \cup 01)$ produces

$$\{1, 10\}\{1, 01\} = \{11, 101, 101, 1001\}.$$

The string 101 is produced in two different ways.

Definition. A regular expression is *unambiguous* if it produces each string in only one way; otherwise it is *ambiguous*.

Example. The expression $(0 \cup 1)^*$ is unambiguous. The expression $(\varepsilon \cup 0 \cup 1)^*$ is ambiguous, since $1 = \varepsilon 1 = 1\varepsilon = \cdots$.

To show that an expression is unambiguous, use the following rules:

- ε, 0, and 1 are unambiguous.
- Suppose R and S are unambiguous and produce the languages $\mathcal{R}$ and $\mathcal{S}$. Then $R \cup S$ is unambiguous if and only if $\mathcal{R} \cap \mathcal{S} = \varnothing$.
- Under the same assumption, RS is unambiguous if and only if every string in $\mathcal{RS}$ has a unique factorization rs with $r \in \mathcal{R}$ and $s \in \mathcal{S}$.
- R^* is unambiguous if and only if each R^k is unambiguous and the languages $\mathcal{R}^k$ are pairwise disjoint.

Lemma. The expression $0^*(10^*)^*$ produces all binary strings and is unambiguous.

Proof. For example, consider the following string, broken at its 1s:

$$\boxed{0\ \ 0}\boxed{1\ \ 0}\boxed{1}\boxed{1\ \ 0}\boxed{1}$$

A string begins with a possibly empty sequence of 0s, then breaks into pieces beginning with 1. We can uniquely decompose it using the locations of its 1s as

$$\underbrace{(0\cdots0)}_{m_0}\underbrace{(10\cdots0)}_{1+m_1}\cdots\underbrace{(10\cdots0)}_{1+m_r},$$

for some $r \geq 0$ and $m_0, m_1, \ldots, m_r \geq 0$. Ranging over all these values gives every binary string, with a unique decomposition.

Corollary. The expression $1^*(01^*)^*$ unambiguously produces all binary strings.

These expressions can be modified to produce strings with restrictions.

Example. Give an unambiguous regular expression for the binary strings in which every 0 is followed by at least one 1.

Start with $1^*(01^*)^*$. In this expression a 0 may be followed by any number of 1s, including none. Requiring at least one gives

$$1^*(011^*)^*.$$

This expression is unambiguous and has generating function

$$\frac{1}{1-x}\cdot\frac{1}{1-\frac{x^2}{1-x}}.$$

Unambiguous expressions translate into generating functions:

Expression	Generating function
ε	1
0 or 1	x
$R \cup S$	$R(x) + S(x)$
RS	$R(x)S(x)$
R^*	$\dfrac{1}{1-R(x)}$

Remark. Unambiguity is the key: union must not count a string twice, concatenation must correspond to the Cartesian product without collisions, and an unambiguous starred expression cannot have the empty string in its repeated language.

4.3 Blocks and restrictions

Definition. A *block* of a binary string is a nonempty maximal consecutive sequence of the same digit.

Example. The string 00011101100 has blocks

$$\underbrace{000}_{3}\ \underbrace{111}_{3}\ \underbrace{0}_{1}\ \underbrace{11}_{2}\ \underbrace{00}_{2}.$$

Theorem. The expressions

$$0^*(11^*00^*)^*1^* \qquad \text{and} \qquad 1^*(00^*11^*)^*0^*$$

unambiguously produce all binary strings.

The idea is to separate the initial possibly empty block, the alternating nonempty blocks, and the final possibly empty block. For example, the first expression starts with 0s (or nothing), then has pairs of a 1-block and a 0-block, and can end in 1s. The maximal blocks determine the decomposition uniquely.

Example. Find the generating function for binary strings in which every block has length at most 2.

Modify $1^*(00^*11^*)^*0^*$ by restricting each block to obtain the unambiguous expression

$$(\varepsilon \cup 1 \cup 11)((0 \cup 00)(1 \cup 11))^*(\varepsilon \cup 0 \cup 00).$$

Its generating function is

$$\begin{aligned} B(x) &= (1+x+x^2)\frac{1}{1-(x+x^2)^2}(1+x+x^2) \\ &= \frac{(1+x+x^2)^2}{(1-x-x^2)(1+x+x^2)} \\ &= \frac{1+x+x^2}{1-x-x^2}. \end{aligned}$$

Example. Find the generating function for binary strings in which every block of 0s has even length.

A nonempty even-length block of 0s is represented unambiguously by $(00)(00)^*$. Thus the required strings are produced by

$$1^*\Big((00)(00)^*11^*\Big)^*(00)^*.$$

The generating function is

$$\begin{aligned} S(x) &= \frac{1}{1-x}\cdot\frac{1}{1-\frac{x^3}{(1-x)(1-x^2)}}\cdot\frac{1}{1-x^2} \\ &= \frac{1}{1-x-x^2}. \end{aligned}$$

Example. Find the generating function for binary strings that do not contain 011 as a substring.

Starting with $1^*(00^*11^*)^*0^*$, restrict the 1-block after any 0-block to have length at most one. This gives

$$1^*(00^*1)^*0^*.$$

The generating function is

$$\frac{1}{1-x} \cdot \frac{1}{1-\frac{x^2}{1-x}} \cdot \frac{1}{1-x} = \frac{1}{1-2x+x^3}.$$

5 Bivariate generating functions

A generating function tracks objects by size. To track other properties, we use more variables.

Definition. A *combinatorial class with a parameter* $(C, |\cdot|, \chi)$ consists of

- a set of objects C;
- a size function $|\cdot| \colon C \to \mathbb{N}$ such that there are finitely many objects of each size;
- a parameter function $\chi \colon C \to \mathbb{N}$.

Its *bivariate generating function* is

$$C(u,z) = \sum_{\sigma \in C} u^{\chi(\sigma)} z^{|\sigma|} = \sum_{n \geq 0} \left(\sum_{k \geq 0} c_{k,n} u^k \right) z^n,$$

where $c_{k,n}$ is the number of objects in C with size n and parameter value k.

Setting $u = 1$ forgets the parameter:

$$C(1,z) = \sum_{\sigma \in C} z^{|\sigma|} = C(z).$$

Example. Let B be the class of binary strings with size given by length and parameter given by the number of 0s. Then

$$\begin{aligned} B(u,z) &= \sum_{n \geq 0} \left(\sum_{k \geq 0} \binom{n}{k} u^k \right) z^n \\ &= \sum_{n \geq 0} (1+u)^n z^n \\ &= \sum_{n \geq 0} \Big((1+u)z \Big)^n = \frac{1}{1-(1+u)z}. \end{aligned}$$

This calculation starts with the counting series. We also want a way to pass directly from classes to generating functions.

5.1 Markers

To build bivariate generating functions, add a *marker* μ to the constructions.

- μ has size 0 and parameter value 1, so its generating function is $z^0u^1 = u$.
- $\mathcal{E}$ has size 0 and parameter value 0, so its generating function is 1.
- $\mathcal{Z}$ has size 1 and parameter value 0, so its generating function is z.

The marker keeps track of the parameter without contributing to size.

Example. For binary strings enumerated by length and number of zeros,

$$B = \mathrm{SEQ}(\mathcal{Z}_0 \times \mu + \mathcal{Z}_1).$$

Thus

$$B(u,z) = \frac{1}{1-uz-z} = \frac{1}{1-(1+u)z}.$$

Example. Let C be the class of integer compositions, tracking the number of parts equal to 1. Its specification is

$$C = \mathrm{SEQ}\Big(\mathcal{Z} \times \mu + \mathcal{Z}^2 \times \mathrm{SEQ}(\mathcal{Z})\Big).$$

The first term is a marked part of size 1; the second represents parts of size at least 2. Hence

$$\begin{aligned} C(u,z) &= \frac{1}{1-\left(zu + \frac{z^2}{1-z}\right)} \\ &= \frac{1-z}{1-(1+u)z+(u-1)z^2}. \end{aligned}$$

This is a two-variable rational generating function.

Example. Find the generating function for rooted planar binary trees enumerated by their number of vertices and number of leaves.

Let T be the class of nonempty binary trees. Then

$$T = \mathcal{Z} + \mathcal{Z} \times \mathcal{E} \times T + \mathcal{Z} \times T \times \mathcal{E} + \mathcal{Z} \times T \times T.$$

These cases correspond to no children, an empty left subtree and nonempty right subtree, a nonempty left subtree and empty right subtree, and two nonempty subtrees. Marking leaves gives

$$T = \mu \times \mathcal{Z} + \mathcal{Z} \times \mathcal{E} \times T + \mathcal{Z} \times T \times \mathcal{E} + \mathcal{Z} \times T \times T,$$

and therefore

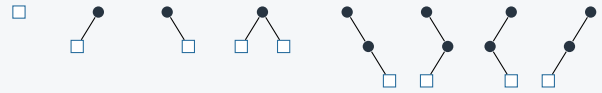
$$T(u,z) = uz + 2zT(u,z) + zT(u,z)^2.$$

Solving the quadratic gives

$$T(u,z) = \frac{1-2z-\sqrt{1-4z+4(1-u)z^2}}{2z}.$$

The other sign introduces a $1/z$ term, so the minus sign is the formal power series solution. Setting $u = 1$ recovers the ordinary generating function for nonempty binary trees. The first terms are

$$T(u,z) = uz + 2uz^2 + (u^2+4u)z^3 + \cdots.$$



uz $\quad 2uz^2$ $\quad (u^2+4u)z^3$

5.2 Expected values

Given $C(u,z) = \sum_{n\geq 0}\left(\sum_{k\geq 0} c_{k,n}u^k\right)z^n$, its formal partial derivative with respect to u is

$$\frac{\partial}{\partial u}C(u,z) = C_u(u,z) = \sum_{n\geq 0}\left(\sum_{k\geq 1} kc_{k,n}u^{k-1}\right)z^n.$$

The *expected value*, or average, of the parameter $\chi\colon C \to \mathbb{N}$ among the objects of size n is

$$\begin{aligned}\mathbb{E}_n[\chi] &= \sum_{k\geq 0} k\,\mathbb{P}\Big(\chi(\sigma) = k \mid |\sigma| = n\Big)\\ &= \sum_{k\geq 0} k\,\frac{\#\{\sigma : \chi(\sigma) = k,\ |\sigma| = n\}}{\#\{\sigma : |\sigma| = n\}}\\ &= \sum_{k\geq 0} k\frac{c_{k,n}}{c_n}.\end{aligned}$$

Here an object of size n is chosen uniformly at random.

Proposition. For each n with $c_n \neq 0$,

$$\mathbb{E}_n[\chi] = \frac{[z^n]C_u(1,z)}{[z^n]C(1,z)}.$$

Proof. Since

$$[z^n]C_u(u,z) = \sum_{k\geq 1} kc_{k,n}u^{k-1}, \qquad [z^n]C_u(1,z) = \sum_{k\geq 0} kc_{k,n},$$

dividing by $[z^n]C(1,z) = c_n$ gives the expected value.

Example. Find the average number of 0s among binary strings of length n.

We know that $B(u,z) = \dfrac{1}{1-(1+u)z}$, so

$$B_u(1,z) = \left.\frac{z}{(1-(1+u)z)^2}\right|_{u=1} = \frac{z}{(1-2z)^2}.$$

For $n \geq 1$, the negative binomial theorem gives

$$\begin{aligned}[z^n]\frac{z}{(1-2z)^2} &= [z^{n-1}](1-2z)^{-2} \\ &= 2^{n-1}[z^{n-1}](1-z)^{-2} \\ &= 2^{n-1}\binom{n}{n-1} = n2^{n-1}.\end{aligned}$$

Thus the expected value is $\dfrac{n2^{n-1}}{2^n} = \dfrac{n}{2}$. For $n = 0$, it is also 0.

Example. The generating function for compositions enumerated by size and number of parts equal to 1 is

$$C(u,z) = \frac{1-z}{(1-z)(1-uz)-z^2}, \qquad [z^n]C(1,z) = 2^{n-1} \quad (n \geq 1).$$

Furthermore,

$$\begin{aligned}C_u(1,z) &= \left.\frac{(1-z)^2 z}{\Big((1-z)(1-uz)-z^2\Big)^2}\right|_{u=1} \\ &= \frac{(1-z)^2 z}{(1-2z)^2}.\end{aligned}$$

For $n \geq 3$,

$$\begin{aligned}[z^n]C_u(1,z) &= [z^n]\frac{z-2z^2+z^3}{(1-2z)^2} \\ &= [z^{n-1}](1-2z)^{-2} - 2[z^{n-2}](1-2z)^{-2} + [z^{n-3}](1-2z)^{-2} \\ &= 2^{n-2} + n2^{n-3}.\end{aligned}$$

Thus, for $n \geq 3$, the average number of parts equal to 1 is

$$\frac{2^{n-2}+n2^{n-3}}{2^{n-1}} = \frac{n+2}{4}.$$

For $n = 0, 1, 2$, the averages are respectively 0, 1, and 1.

5.3 Variance and concentration

The *variance* of the parameter χ on objects of size n is

$$\mathrm{Var}_n[\chi] = \mathbb{E}_n[\chi^2] - \mathbb{E}_n[\chi]^2, \qquad \mathbb{E}_n[\chi^2] = \sum_{k\geq 0} k^2 \frac{c_{k,n}}{c_n}.$$

The second derivative contributes $k(k-1)$, so adding the first derivative gives the factor k^2.

Proposition. For $c_n \neq 0$,

$$\mathrm{Var}_n[\chi] = \frac{[z^n]\big(C_{uu}(1,z) + C_u(1,z)\big)}{[z^n]C(1,z)} - \left(\frac{[z^n]C_u(1,z)}{[z^n]C(1,z)}\right)^2.$$

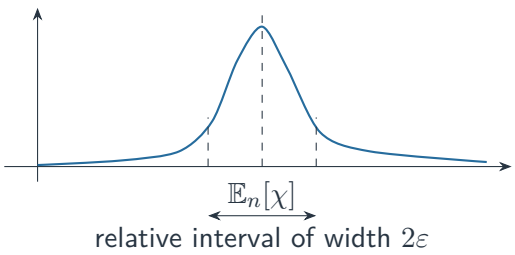
Proposition. Suppose $\mathbb{E}_n[\chi] \neq 0$ for all sufficiently large n, and

$$\lim_{n\to\infty} \frac{\mathrm{Var}_n[\chi]}{\mathbb{E}_n[\chi]^2} = 0.$$

Then, for every $\varepsilon > 0$, the probability that a uniformly random object $\sigma \in C$ of size n satisfies

$$(1-\varepsilon)\mathbb{E}_n[\chi] \leq \chi(\sigma) \leq (1+\varepsilon)\mathbb{E}_n[\chi]$$

tends to 1 as $n \to \infty$.



6 The Lagrange implicit function theorem

Theorem (Lagrange implicit function theorem (LIFT))**.** Let R be a power series in z with no constant term, satisfying

$$z = \frac{R(z)}{G(R(z))},$$

where $G(t)$ is a series such that $1/G(t)$ is a power series. Then, for $n \geq 1$,

$$[z^n]R(z) = \frac{1}{n}[t^{n-1}]G(t)^n.$$

More generally, for positive integers k,

$$[z^n]R(z)^k = \frac{k}{n}[t^{n-k}]G(t)^n.$$

Example. The class T of nonempty rooted planar trees consists of a root vertex followed by some, possibly no, ordered subtrees. It satisfies

$$T = \mathcal{Z} \times \mathrm{SEQ}(T).$$

Thus

$$T(z) = \frac{z}{1 - T(z)}, \qquad z = \frac{T(z)}{G(T(z))}, \qquad G(t) = (1-t)^{-1}.$$

Therefore, for $n \geq 1$,

$$\begin{aligned}
[z^n]T(z) &= \frac{1}{n}[t^{n-1}](1-t)^{-n} && \text{(LIFT)} \\
&= \frac{1}{n}(-1)^{n-1}\binom{-n}{n-1} && \text{(negative binomial theorem)} \\
&= \frac{1}{n}\binom{2n-2}{n-1}.
\end{aligned}$$

For comparison, $\dfrac{1}{n+1}\binom{2n}{n}$ counts rooted planar binary trees with n vertices.

Example. Fix a positive integer r. Find the number of rooted planar trees in which every vertex has either 0 or exactly r nonempty subtrees.

The combinatorial description implies that the class T of these nonempty trees satisfies

$$T = \mathcal{Z} + \mathcal{Z} \times T^r,$$

so

$$T(z) = z + zT(z)^r = z\Big(1 + T(z)^r\Big).$$

Therefore

$$z = \frac{T(z)}{G(T(z))}, \qquad G(t) = 1 + t^r.$$

LIFT and the binomial theorem imply

$$\begin{aligned}
[z^n]T(z) &= \frac{1}{n}[t^{n-1}](1+t^r)^n \\
&= \begin{cases} 0, & r \nmid n-1, \\ \dfrac{1}{n}\dbinom{n}{(n-1)/r}, & r \mid n-1. \end{cases}
\end{aligned}$$

In the second case, $n - 1 = kr$ for an integer k.

Example. The class of rooted planar 5-ary trees satisfies

$$F = \mathcal{E} + \mathcal{Z} \times F^5, \qquad F(z) = 1 + zF(z)^5.$$

To apply LIFT, we first need to remove the constant term. Let $S(z) = F(z) - 1$. Then

$$S(z) = z\Big(S(z) + 1\Big)^5, \qquad z = \frac{S(z)}{G(S(z))}, \qquad G(t) = (1+t)^5.$$

By the Lagrange inversion formula, for $n \geq 1$,

$$\begin{aligned}[z^n]F(z) &= [z^n]S(z) \\ &= \frac{1}{n}[t^{n-1}](1+t)^{5n} \\ &= \frac{1}{n}\binom{5n}{n-1}.\end{aligned}$$

Therefore

$$F(z) = 1 + \sum_{n\geq 1} \frac{1}{n}\binom{5n}{n-1} z^n.$$

6.1 Expected number of leaves using Lagrange inversion

Suppose $F(u,z)$ is a bivariate generating function, with parameter χ. Recall that

$$\mathbb{E}_n[\chi] = \frac{[z^n]\left.\dfrac{\partial}{\partial u}F(u,z)\right|_{u=1}}{[z^n]F(1,z)}.$$

Coefficient extraction and differentiation commute:

$$[z^n]\left.\frac{\partial}{\partial u}F(u,z)\right|_{u=1} = \left.\frac{\partial}{\partial u}[z^n]F(u,z)\right|_{u=1}.$$

Example. The bivariate generating function for nonempty binary trees, enumerated by size and number of leaves, satisfies

$$N(u,z) = uz + 2zN(u,z) + zN(u,z)^2.$$

Here the uz term represents a single leaf. Thus

$$z = \frac{N(u,z)}{G_u(N(u,z))}, \qquad G_u(t) = u + 2t + t^2.$$

Using Lagrange inversion,

$$\begin{aligned}[z^n]\left.\frac{\partial}{\partial u}N(u,z)\right|_{u=1} &= \left.\frac{\partial}{\partial u}[z^n]N(u,z)\right|_{u=1} \\ &= \left.\frac{\partial}{\partial u}\left(\frac{1}{n}[t^{n-1}](u+2t+t^2)^n\right)\right|_{u=1} \\ &= \frac{1}{n}[t^{n-1}]\left.\frac{\partial}{\partial u}(u+2t+t^2)^n\right|_{u=1} \\ &= \frac{1}{n}[t^{n-1}]\left.n(u+2t+t^2)^{n-1}\right|_{u=1} \\ &= [t^{n-1}](1+2t+t^2)^{n-1} \\ &= [t^{n-1}](1+t)^{2(n-1)} = \binom{2n-2}{n-1}.\end{aligned}$$

Consequently, for $n \geq 1$,

$$\begin{aligned}\mathbb{E}_n[\chi] &= \frac{\binom{2n-2}{n-1}}{\frac{1}{n+1}\binom{2n}{n}} \\ &= (n+1)\frac{(2n-2)!}{(2n)!}\frac{n!}{(n-1)!}\frac{n!}{(n-1)!} \\ &= \frac{n(n+1)}{2(2n-1)} \sim \frac{n}{4}.\end{aligned}$$

Thus asymptotically one quarter of the vertices in a uniformly chosen binary tree of size n are leaves, on average.

7 Rational generating functions

Definition. A *C-finite sequence* $(f_n) = (f_0, f_1, \ldots)$ is a sequence satisfying a linear recurrence

$$f_n + c_1 f_{n-1} + c_2 f_{n-2} + \cdots + c_r f_{n-r} = 0 \qquad (n \geq N),$$

where $c_i \in \mathbb{Q}$, $c_r \neq 0$, and $N \geq r$. The integer r is the *order* of the recurrence.

The solutions of a fixed recurrence are closed under addition and scalar multiplication, so they form a vector space. A particular solution is distinguished by its initial conditions $f_0, f_1, \ldots, f_{N-1}$.

Example. The Fibonacci numbers satisfy

$$f_{n+2} = f_{n+1} + f_n \qquad (n \geq 0), \qquad f_0 = f_1 = 1.$$

Equivalently, $f_n - f_{n-1} - f_{n-2} = 0$ for all $n \geq 2$.

Theorem. The sequence (f_n) satisfies the recurrence above if and only if

$$F(z) = \sum_{n \geq 0} f_n z^n = \frac{G(z)}{H(z)},$$

where

$$G(z) = g_0 + g_1 z + \cdots + g_{N-1} z^{N-1}, \qquad H(z) = 1 + c_1 z + \cdots + c_r z^r,$$

with $g_k = f_k + \sum_{i=1}^{k} c_i f_{k-i}$, taking $c_i = 0$ for $i > r$.

Proof. Define $c_0 = 1$, so $H(z) = c_0 + c_1 z + \cdots + c_r z^r$. Then

$$H(z)F(z) = \left(\sum_{k=0}^{r} c_k z^k\right)\left(\sum_{k \geq 0} f_k z^k\right)$$

$$= \sum_{n \geq 0} \left(\sum_{k=0}^{\min(r,n)} c_k f_{n-k} \right) z^n.$$

The recurrence holds exactly when these coefficients vanish for $n \geq N$, that is, when $H(z)F(z)$ is the polynomial $G(z)$ of degree at most $N-1$.

Example. The generating function of binary strings with no blocks longer than k is

$$W(z) = \frac{1 - z^{k+1}}{1 - 2z + z^{k+1}}.$$

Thus, by comparing coefficients after multiplying by the denominator,

$$w_n = 2w_{n-1} - w_{n-k-1} \qquad (n \geq k+2).$$

Example. Consider the look-and-say sequence

$$1,\ 11,\ 21,\ 1211,\ 111221, \ldots.$$

Index the initial term 1 by $n = 0$, and let d_n be the number of digits in the nth term. Its rational generating function is given in abbreviated form by

$$D(z) = \sum_{n \geq 0} d_n z^n = \frac{1 + z + \cdots + 18z^{77} - 12z^{78}}{1 - z + \cdots - 9z^{71} + 6z^{72}}.$$

The numerator has degree 78 and the denominator has degree 72. This illustrates that even a sequence with a simple rule can have a complicated rational generating function. The formula counts digits, rather than giving the look-and-say terms themselves.

Corollary. If $(f_n)_{n \geq 0}$ and $(g_n)_{n \geq 0}$ are C-finite, then their sum and convolution are also C-finite sequences:

$$(f_n + g_n)_{n \geq 0}, \qquad (h_n)_{n \geq 0}, \quad h_n = \sum_{k=0}^{n} f_k g_{n-k}.$$

7.1 The C-finite coefficient theorem

7.1.1 Polynomial roots and multiplicities

The fundamental theorem of algebra states that any polynomial $P(x) \in \mathbb{C}[x]$ of degree d can be factored as

$$P(x) = c(x - \lambda_1) \cdots (x - \lambda_d),$$

for $c, \lambda_1, \ldots, \lambda_d \in \mathbb{C}$, with $c \neq 0$. Grouping repeated roots gives

$$P(x) = c(x - \lambda_1)^{d_1} \cdots (x - \lambda_s)^{d_s},$$

where the λ_i are distinct. The *multiplicity* of λ_i is d_i. If $d_i = 1$, then λ_i is a *simple root*.

Lemma. The multiplicity of a root λ_i is the smallest positive integer k such that

$$P(\lambda_i) = P'(\lambda_i) = \cdots = P^{(k-1)}(\lambda_i) = 0, \qquad P^{(k)}(\lambda_i) \neq 0.$$

Proof. If $P(x) = c(x-\lambda_1)^{d_1} \cdots (x-\lambda_s)^{d_s}$, then the product rule writes $P^{(j)}(x)$ as a finite sum of terms. When $j < d_i$, each term retains a factor $x - \lambda_i$, so $P^{(j)}(\lambda_i) = 0$. Also,

$$P^{(d_i)}(\lambda_i) = c\, d_i! \prod_{\ell \neq i} (\lambda_i - \lambda_\ell)^{d_\ell} \neq 0,$$

since the roots are distinct.

7.1.2 Partial fraction decomposition

Suppose $F(z) = G(z)/H(z)$, where $\deg G < \deg H$, $H(0) \neq 0$, and

$$H(z) = c(z-\lambda_1)^{d_1} \cdots (z-\lambda_s)^{d_s},$$

with distinct nonzero complex roots $\lambda_1, \ldots, \lambda_s$. Then there exist complex numbers $C_j^{(i)}$ such that

$$F(z) = \sum_{i=1}^{s} \sum_{j=1}^{d_i} \frac{C_j^{(i)}}{(1 - z/\lambda_i)^j}.$$

Theorem (C-finite coefficient theorem)**.** Suppose (f_n) is a C-finite sequence with rational generating function

$$F(z) = R(z) + \frac{G(z)}{H(z)},$$

where R is a polynomial, $\deg G < \deg H$, and $H(0) \neq 0$. If $\lambda_1, \ldots, \lambda_s$ are the distinct roots of H, with multiplicities $d_1, \ldots, d_s$, then

$$f_n = P_1(n)\lambda_1^{-n} + \cdots + P_s(n)\lambda_s^{-n} \qquad (n > \deg R),$$

where $P_i(n)$ is a polynomial in n of degree at most $d_i - 1$.

Example. For $F(z) = \dfrac{1}{3(z-1)^3(z-2)}$, the coefficient theorem gives a formula of the form

$$f_n = C\left(\frac{1}{2}\right)^n + (An^2 + Bn + D)1^n.$$

Proof. Let $n > \deg R$, so $f_n = [z^n]G(z)/H(z)$. Partial fraction decomposition gives

$$\frac{G(z)}{H(z)} = A_1(z) + \cdots + A_s(z), \qquad A_i(z) = \sum_{j=1}^{d_i} \frac{C_j^{(i)}}{(1 - z/\lambda_i)^j}.$$

For a positive integer m, the negative binomial theorem gives

$$\begin{aligned}[z^n](1-z/\lambda_i)^{-m} &= (-\lambda_i)^{-n}\binom{-m}{n} \\ &= \lambda_i^{-n}\binom{m+n-1}{m-1} \\ &= \lambda_i^{-n}\frac{(n+m-1)(n+m-2)\cdots(n+1)}{(m-1)!}.\end{aligned}$$

Thus $[z^n]A_i(z) = \lambda_i^{-n}P_i(n)$, where $P_i(n)$ is a polynomial of degree at most $d_i - 1$.

Corollary. If (f_n) satisfies

$$f_n + c_1 f_{n-1} + \cdots + c_r f_{n-r} = 0 \qquad (n \geq r),$$

then

$$f_n = P_1(n)\lambda_1^{-n} + \cdots + P_s(n)\lambda_s^{-n} \qquad (n \geq 0),$$

where the λ_i are roots of

$$H(z) = 1 + c_1 z + \cdots + c_r z^r$$

and $\deg P_i \leq$ (multiplicity of λ_i as a root of H) $- 1$.

Remark. The notes call H the characteristic polynomial. In the convention where the characteristic polynomial is $t^r + c_1 t^{r-1} + \cdots + c_r$, its roots are the reciprocals of the λ_i used here.

Example. Find a closed form for the sequence satisfying

$$f_{n+3} = 3f_{n+1} - 2f_n \qquad (n \geq 0), \qquad f_0 = f_1 = 4, \quad f_2 = 13.$$

The denominator polynomial factors as

$$H(z) = 1 - 3z^2 + 2z^3 = (1-z)^2(1+2z).$$

The C-finite coefficient theorem gives constants $A, B, C \in \mathbb{C}$ such that

$$f_n = (A + Bn)1^{-n} + C\left(-\frac{1}{2}\right)^{-n} = A + Bn + C(-2)^n.$$

The initial values give

$$\begin{aligned}4 &= f_0 = A + C, \\ 4 &= f_1 = A + B - 2C, \\ 13 &= f_2 = A + 2B + 4C.\end{aligned}$$

Hence $A = 3$, $B = 3$, $C = 1$, and

$$f_n = 3 + 3n + (-2)^n.$$

7.2 Coefficients from poles

Lemma. Assume the hypotheses of the C-finite coefficient theorem. If λ_k is a simple root of $H(z)$ and $G(\lambda_k) \neq 0$, then the polynomial $P_k(n)$ is the constant

$$P_k(n) = -\frac{G(\lambda_k)}{\lambda_k H'(\lambda_k)}.$$

Proof. Since λ_k is a simple root, write

$$H(z) = (z - \lambda_k)I(z), \qquad I(\lambda_k) \neq 0.$$

Partial fraction decomposition gives

$$\frac{G(z)}{H(z)} = \frac{G(z)}{(z-\lambda_k)I(z)} = \frac{c}{z-\lambda_k} + \frac{J(z)}{I(z)}.$$

Clearing denominators,

$$G(z) = cI(z) + J(z)(z - \lambda_k).$$

Setting $z = \lambda_k$ gives $G(\lambda_k) = cI(\lambda_k)$, hence $c = G(\lambda_k)/I(\lambda_k)$. Also,

$$H'(z) = I(z) + (z-\lambda_k)I'(z), \qquad H'(\lambda_k) = I(\lambda_k).$$

The polynomial $P_k(n)$ is therefore obtained from

$$\begin{aligned}
[z^n]\frac{c}{z-\lambda_k} &= -\frac{c}{\lambda_k}[z^n]\frac{1}{1-z/\lambda_k} \\
&= -\frac{c}{\lambda_k}\lambda_k^{-n} \\
&= -\frac{G(\lambda_k)}{\lambda_k H'(\lambda_k)}\lambda_k^{-n}.
\end{aligned}$$

Lemma. Assume the same hypotheses. If λ_k is a root of $H(z)$ of multiplicity m and $G(\lambda_k) \neq 0$, then

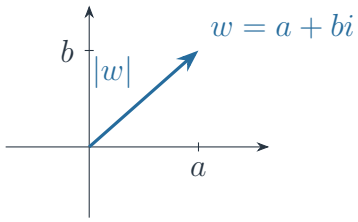
$$P_k(n) = \frac{m(-1)^m G(\lambda_k)}{\lambda_k^m H^{(m)}(\lambda_k)} n^{m-1} + O(n^{m-2}).$$

7.3 Asymptotic notation and dominant roots

For sequences (f_n) and (g_n) of positive numbers, we write

- $f_n \sim g_n$ if $\lim_{n\to\infty} \frac{f_n}{g_n} = 1$;
- $f_n = O(g_n)$ if there are $C > 0$ and $N \in \mathbb{N}$ such that $f_n \leq C g_n$ for all $n \geq N$;
- $f_n = o(g_n)$ if $\lim_{n\to\infty} \frac{f_n}{g_n} = 0$.

For $w \in \mathbb{C}$, write $w = a + bi$.



The *modulus* of w is $|w| = \sqrt{a^2 + b^2}$. For complex sequences, $f_n = O(g_n)$ means $|f_n| = O(|g_n|)$.

A C-finite sequence can be written, for all sufficiently large n, as

$$f_n = P_1(n)\lambda_1^{-n} + \cdots + P_s(n)\lambda_s^{-n}.$$

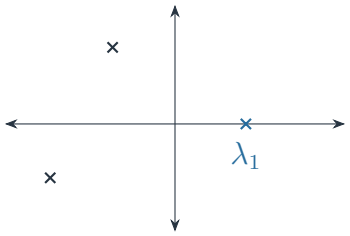
Corollary. Assume the hypotheses of the C-finite coefficient theorem and that G and H are coprime. If

$$|\lambda_1| < |\lambda_k| \qquad (2 \le k \le s),$$

then

$$f_n = P_1(n)\lambda_1^{-n} + O(w^{-n}n^{\alpha-1}),$$

where $w = \min_{2 \le k \le s} |\lambda_k|$, and α is the largest multiplicity among the roots of H with modulus w.



Corollary. Under the same assumptions, if $|\lambda_1| \le |\lambda_k|$ for all k, and λ_1 has multiplicity m greater than that of every other root with the same modulus, then

$$f_n = \frac{m(-1)^m G(\lambda_1)}{\lambda_1^m H^{(m)}(\lambda_1)} \lambda_1^{-n} n^{m-1} \left(1 + O\left(\frac{1}{n}\right)\right).$$

Example. The generating function of integer partitions in which every part is at most 3 is

$$F(x) = \frac{1}{(1-x)(1-x^2)(1-x^3)}.$$

Find the asymptotic number of these partitions as $n \to \infty$.

The denominator factors as

$$H(x) = (1-x)^3(1+x)(1+x+x^2).$$

All its roots have modulus 1, and $x = 1$ is the unique root with highest multiplicity, 3. Here

$G(x) = 1$ does not vanish at $x = 1$. Since $H^{(3)}(1) = -36$,

$$[x^n]F(x) = \left(\frac{-3}{-36}n^2 + O(n)\right) 1^{-n} \sim \frac{n^2}{12}.$$